

OSLO-M-ROW Trouble Shooting Guide

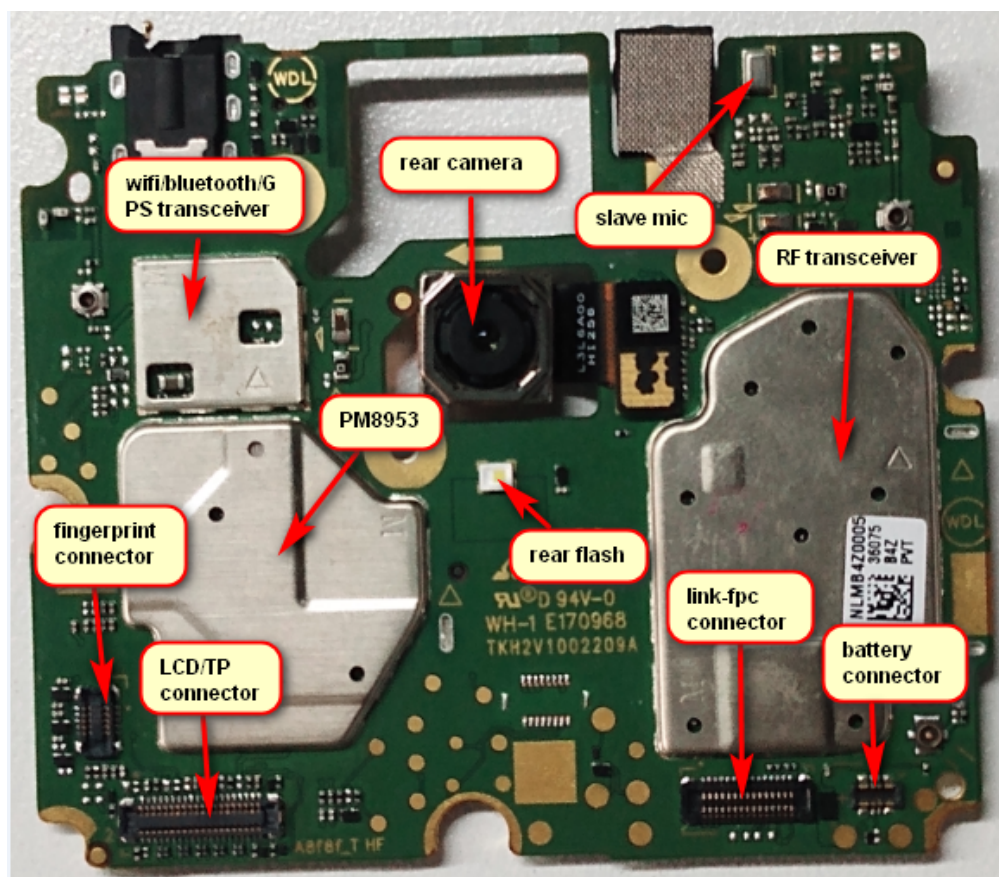
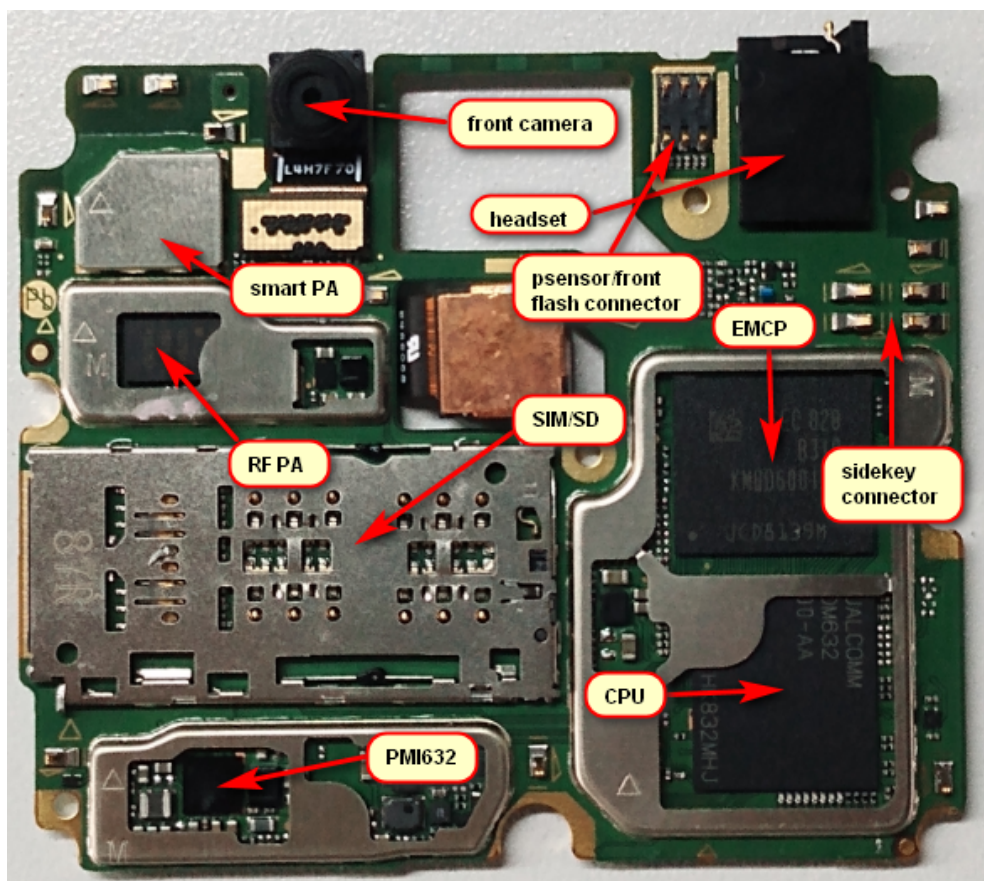
V1.0

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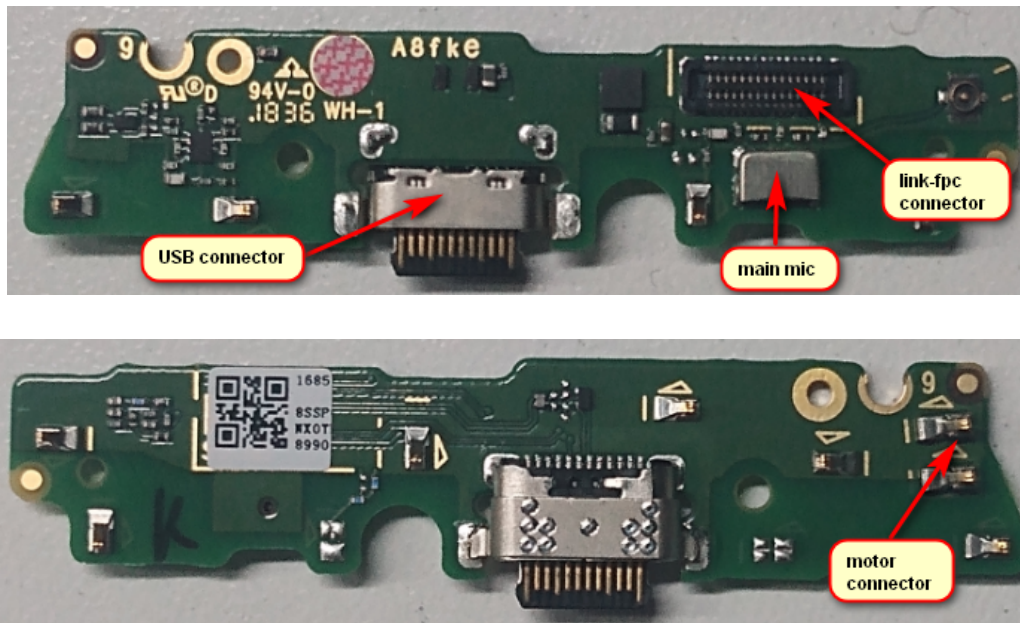
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1. MB/KB illustration



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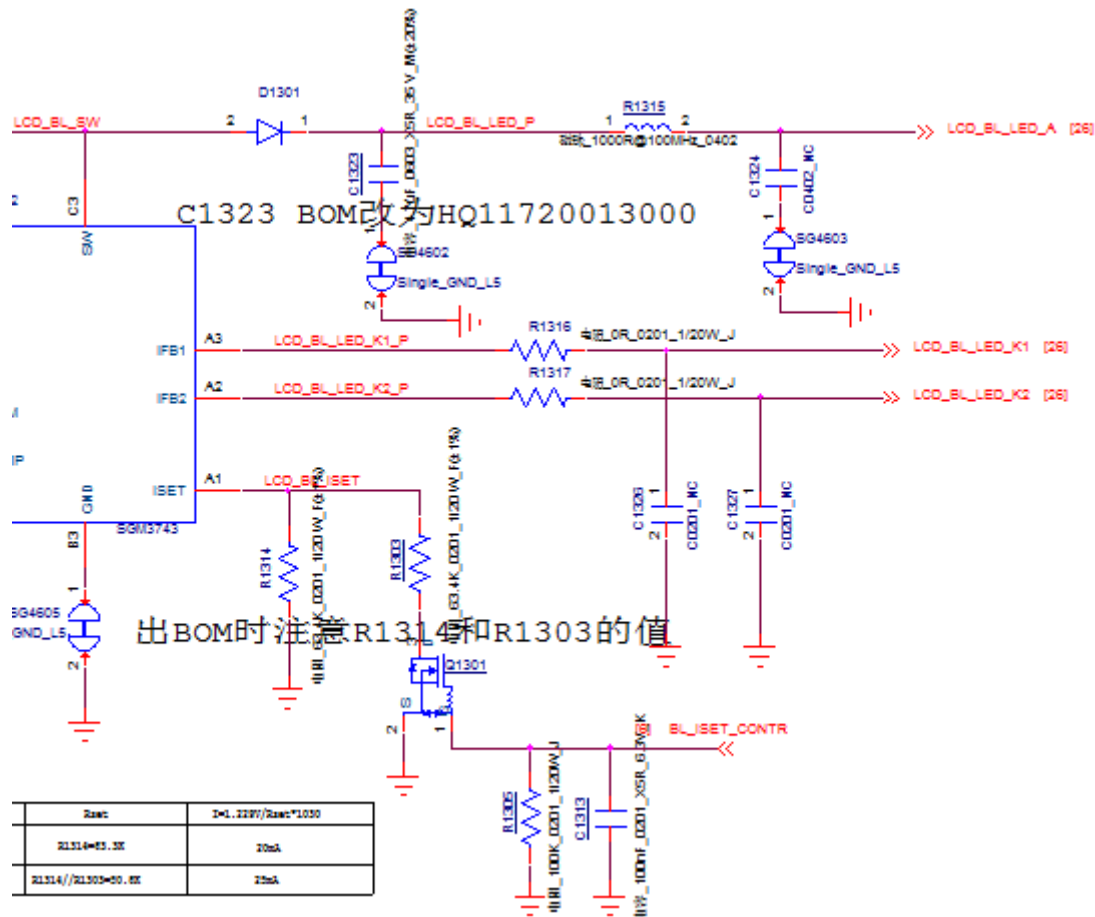
2. Power on failure

Solution:

- 1 Check battery connector for soldering errors and its body for oxidation or damage;
- 2 Check each power supply in the power sequence for short circuits with GND or for any voltage errors;
- 3 Check power on circuit for any errors;
- 4 For power on current around 20MA, check crystal for soldering errors;
- 5 For power on current errors above 1A, check VBAT circuit for short circuits to ground, troubleshoot ground diode at individual locations;
- 6 For power-on current errors above 100MA, check for hot point on MB and troubleshoot;
- 7 Check soldering of each main chip with X-ray, including CPU/Flash/PMIC/TR IC;
- 8 Add-solder or re-solder main chip, validate soldering conditions;
- 9 Replace CPU/Flash/PMIC/LDO/TR IC;

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3. Screen is not bright

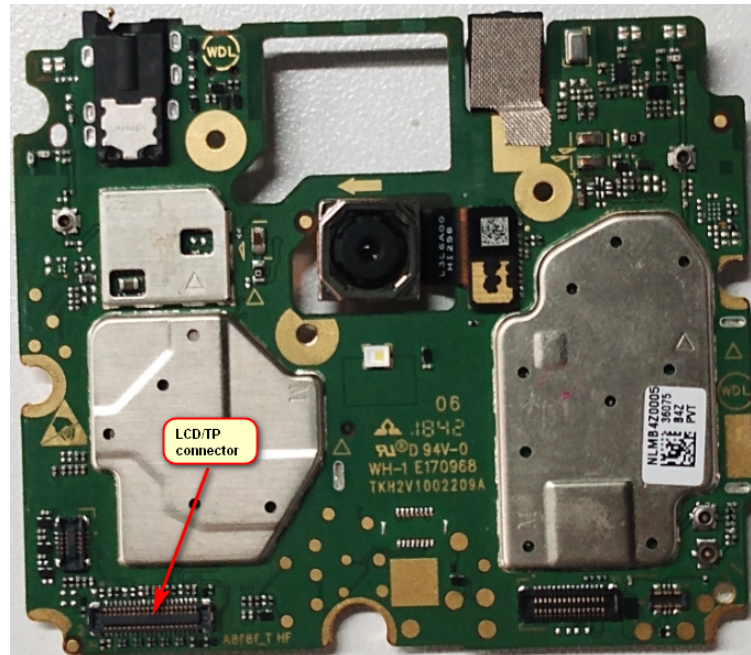


Solution:

- 1 Check LCD connector for dirt, PIN continuous tin, cold solder joint, PIN collapse, etc. Check the connector for proper connection and re-install it;
- 2 LCD backlight powered by U1302, check whether output voltage falls is about 17V;
- 3 Otherwise, check whether R1315、R1316、R1317、D1301、Q1301 is conductive, with soldering error or is damaged;
- 4 Try re-placing D1301;
- 5 Try re-placing D1302;

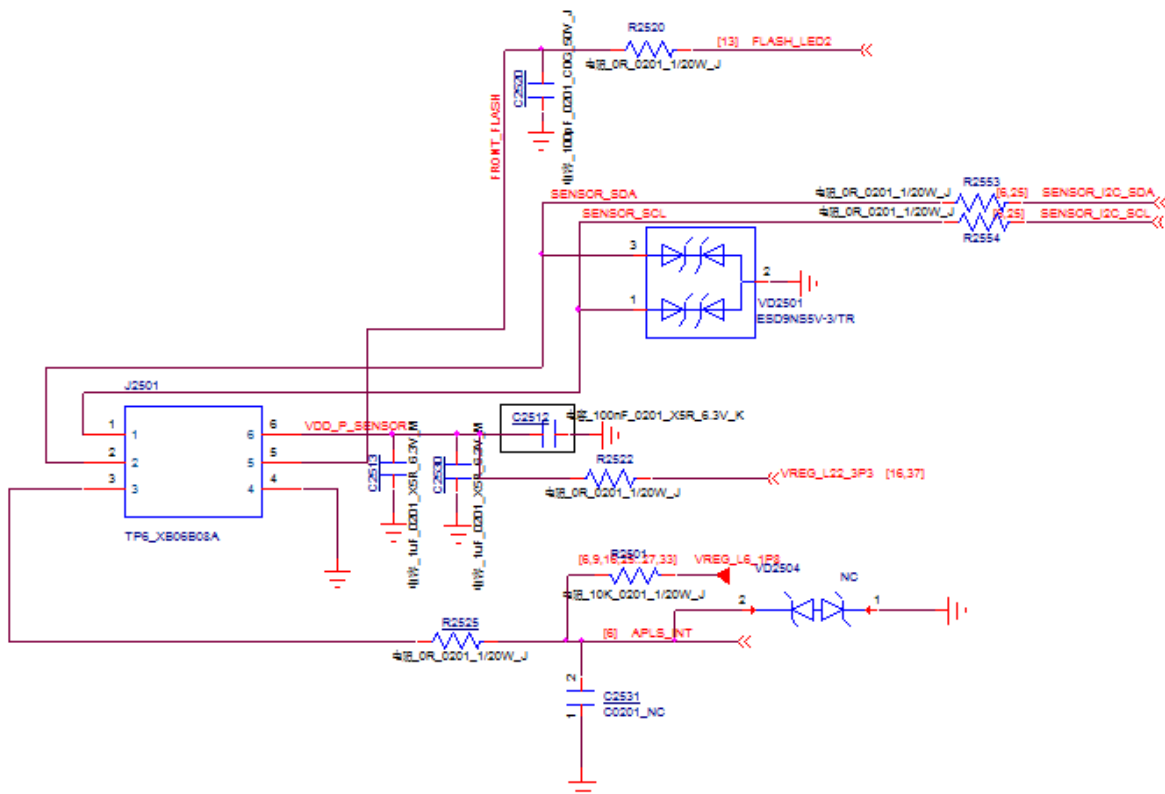
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- 3 Measure TP VDD supply voltage and check whether it is normal. Measure earth impedance of RST, INT, I2C information and check whether it is normal;
- 4 Check the CPU soldering status with an X-ray, add-solder or re-solder it before testing it again;
- 5 Replace CPU;



6. Optical/distance sensor error

ALS/P SENSOR+Front Flash+Charge LED CONN



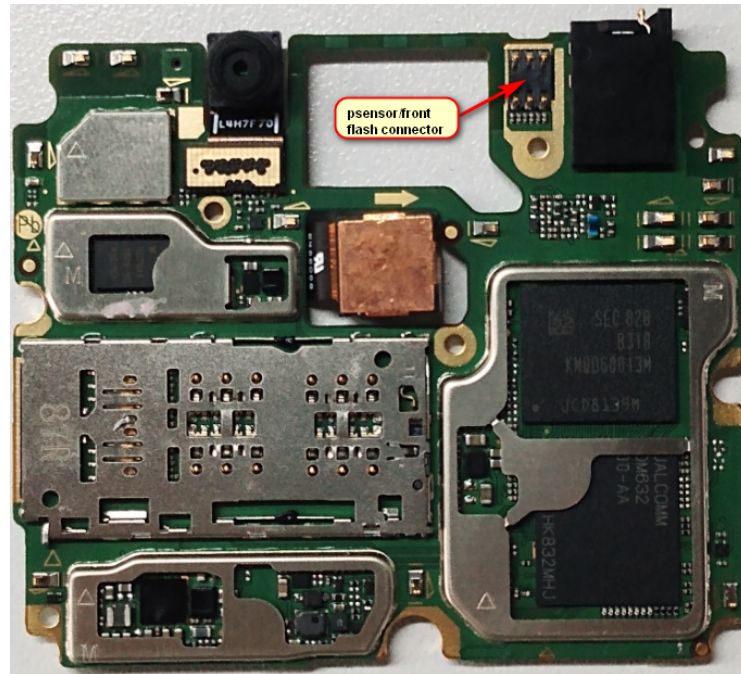
I2C ADDRESS 0X39

Solution:

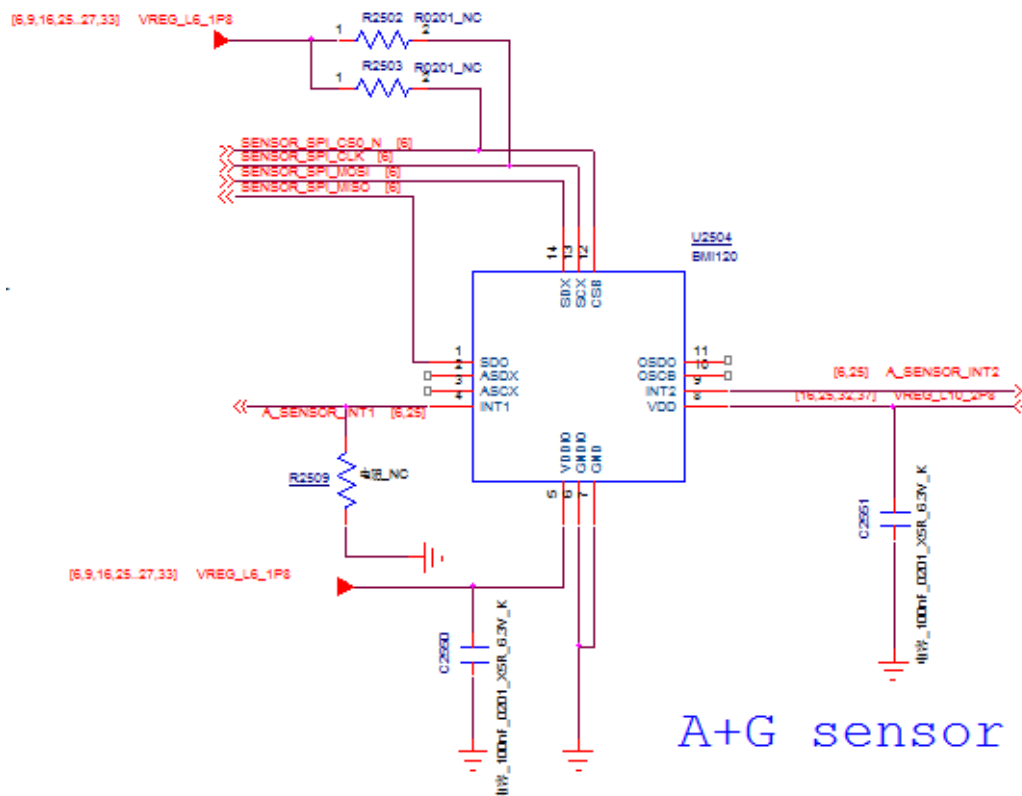
- 1 Check optical chip surface for dirt or blockage; Check whether the plastic cap is well mounted;

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- 2 Check optical/distance sensor for damage or soldering error;
- 3 Measure whether the VREG_L22_3P3 voltage on both ends of R2522 is normal;
- 4 Measure whether the VREG_L6_1P8 voltage on both ends of R2501 is normal;
- 5 Re-place optical and distance sensor FPC;
- 6 Add-solder or re-solder CPU or re-place it;



7. G-sensor and gyroscope failure

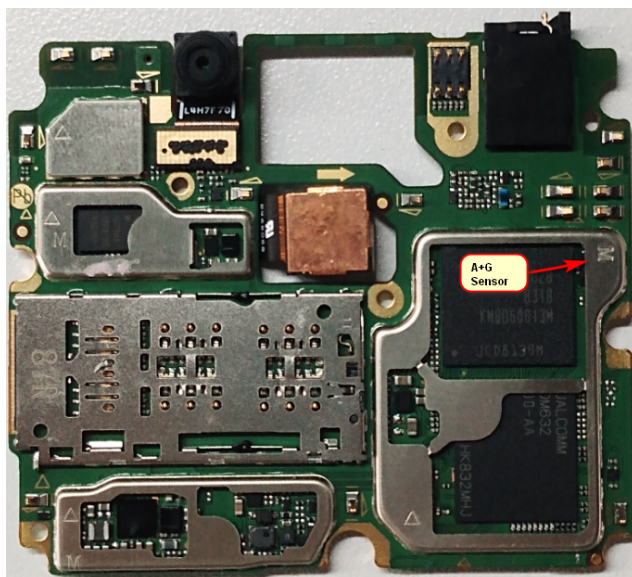


Solution:

- 1 Check the sensor module circuit components for soldering errors or damage;

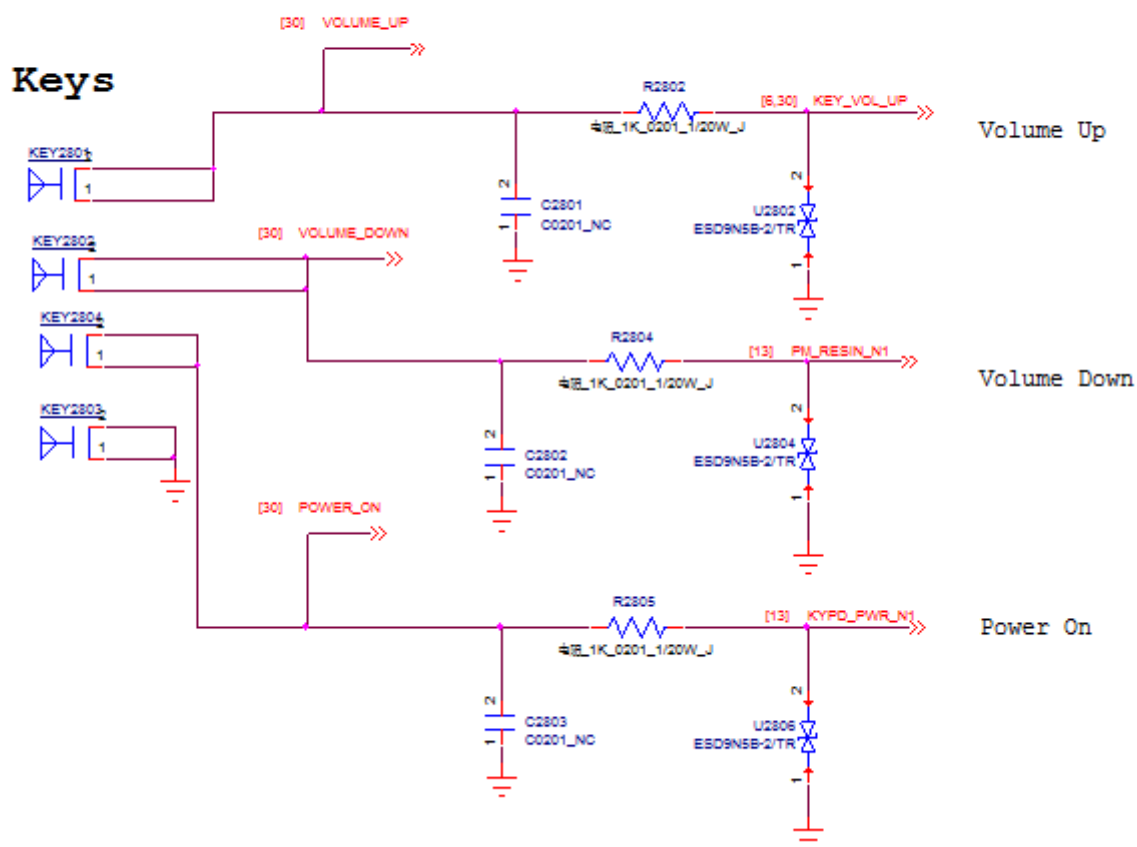
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- 2 Check whether it's functionless or function error;
- 3 Troubleshoot peripheral and channel devices
- 4 Re-solder or re-place the sensor U2504;
- 5 Re-solder or re-place the CPU;



8. Key failure

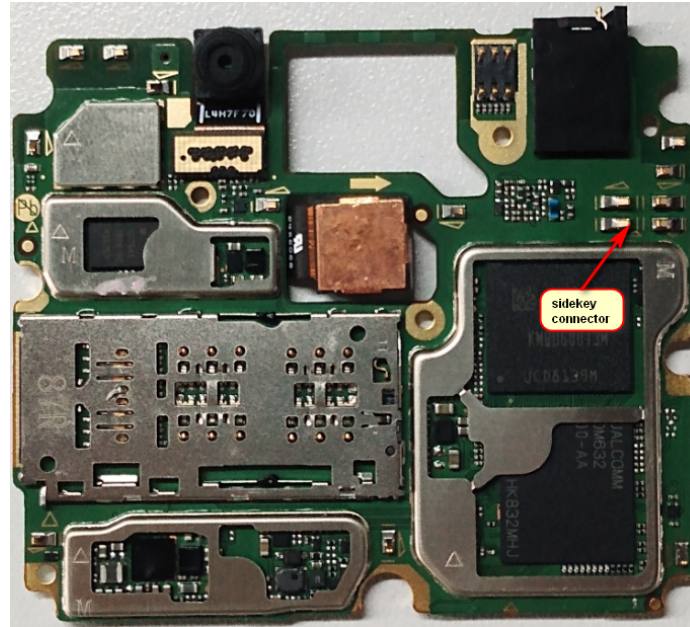
音量键&开关机键



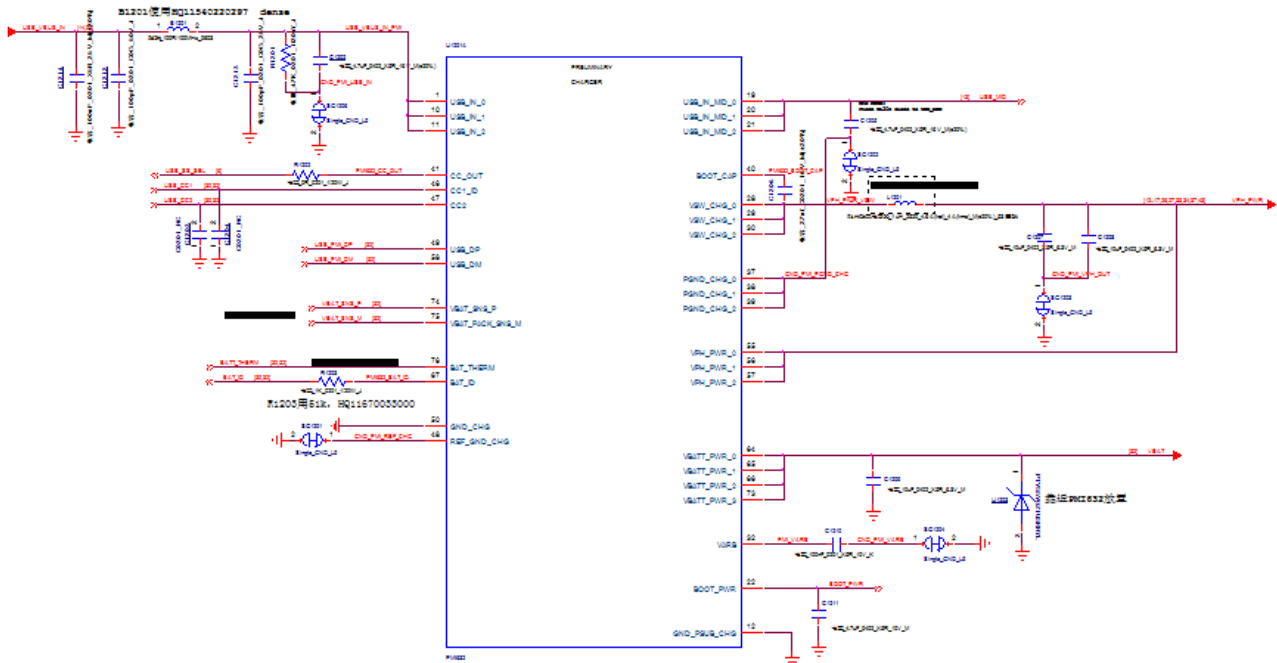
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Solution:

- 1 Check whether the dome falls off or is collapsed, or test the side key FPC after replacing it;
- 2 Check Volume UP/Volume down/Power On voltage signal validity;
- 3 Check components on key circuit for damage or soldering errors;
- 4 Re-solder or replace the CPU or PMIC;



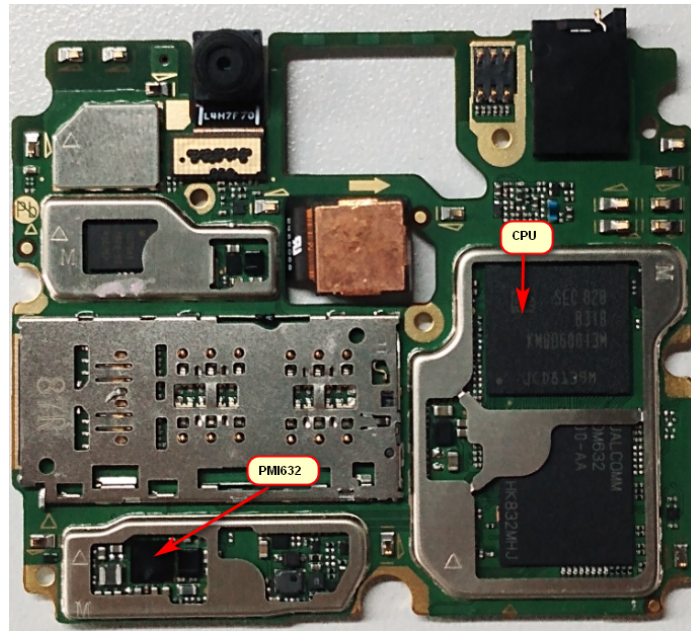
9. No charging



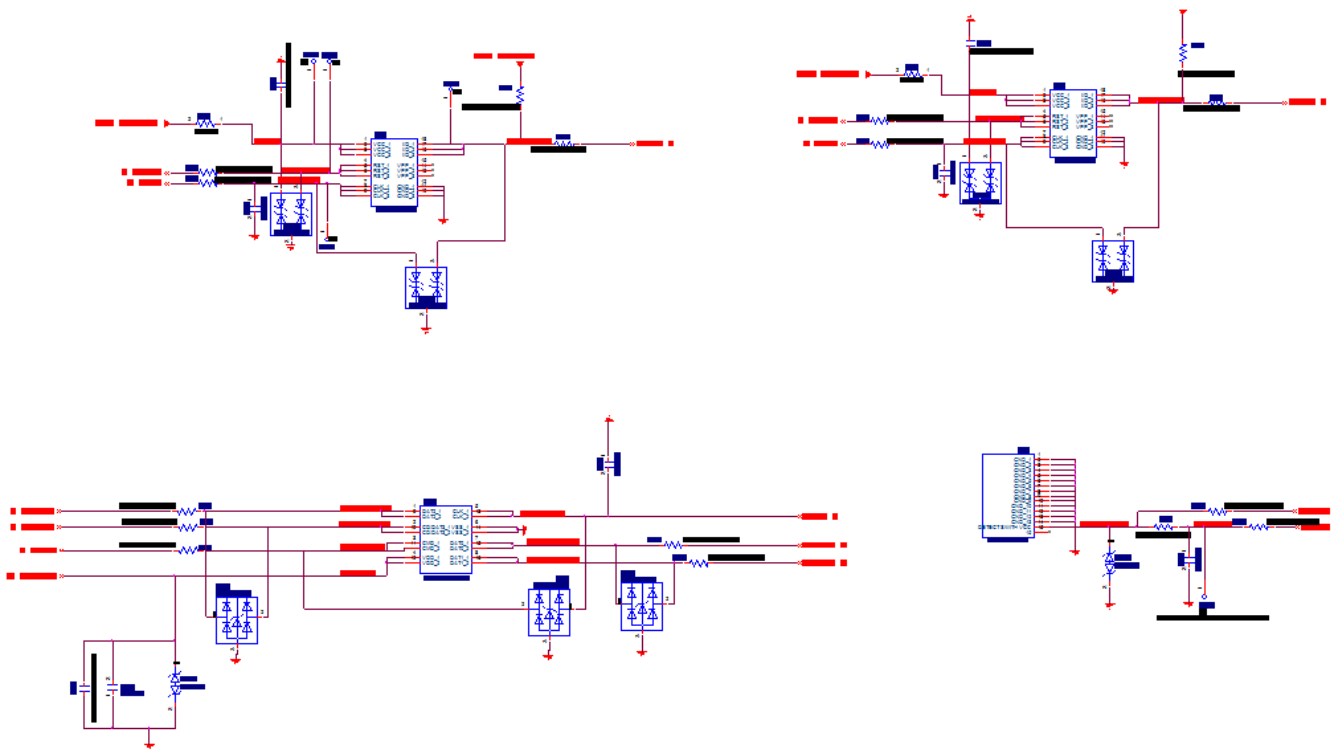
Solution:

- 1 Check USB connector on the motherboard for dirt, damage, cold solder joint or other phenomena;
- 2 Check battery connector J3202 for dirt, damage, bubble soldering;
- 3 Re-place battery before testing again;
- 4 Check whether the voltage on L1201 channel is normal
- 5 Check the power management chip U1201 for soldering errors;

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10. SIM card or SD card identification failure



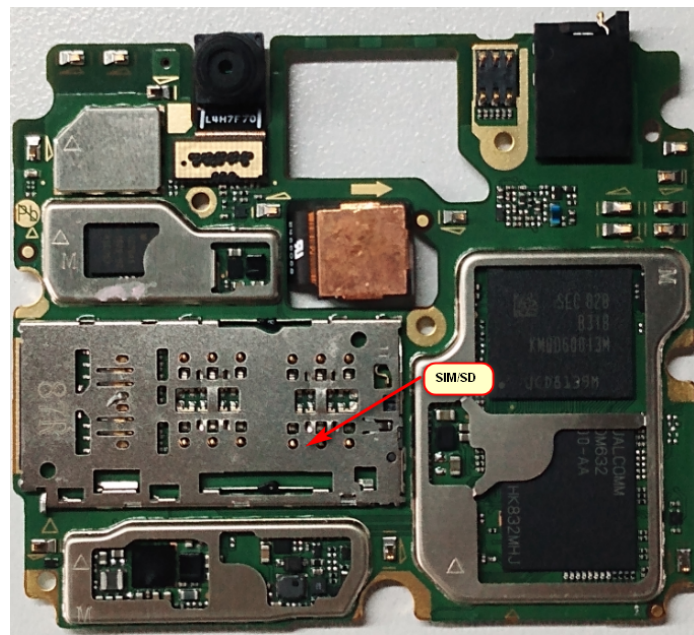
Solution:

- 1 Visually check SIM/SD tray for soldering offset, float or body damage;
- 2 Remove J2903, check whether the DETECT dome positioned correctly (should be connected with metal casing).
- 3 Check SIM/SD tray on the motherboard for float, dome plate contact, spring plate deformation and other errors;
- 4 Measure the SIM power supply voltage VREG_L14_UIM1, Check whether the UIM1, VREG_L15_UIM2 is powering up;
- 5 Check whether the ground resistance of SIM/SD signal wire is normal; Otherwise, disconnect serial

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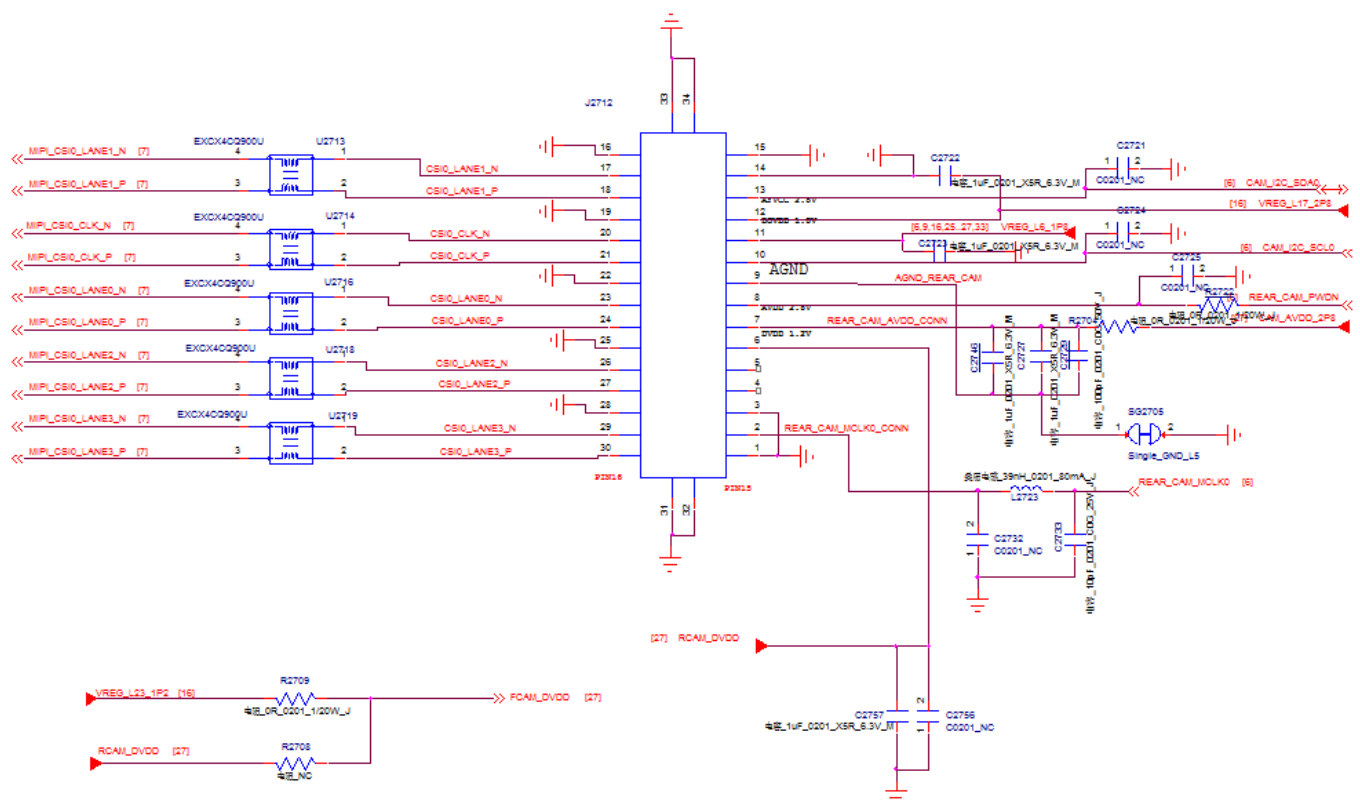
connected resistor and test the CPU side.

- 6 Re-solder or re-place U401 CPU;



11. Main camera function failure

Main Rear Camera



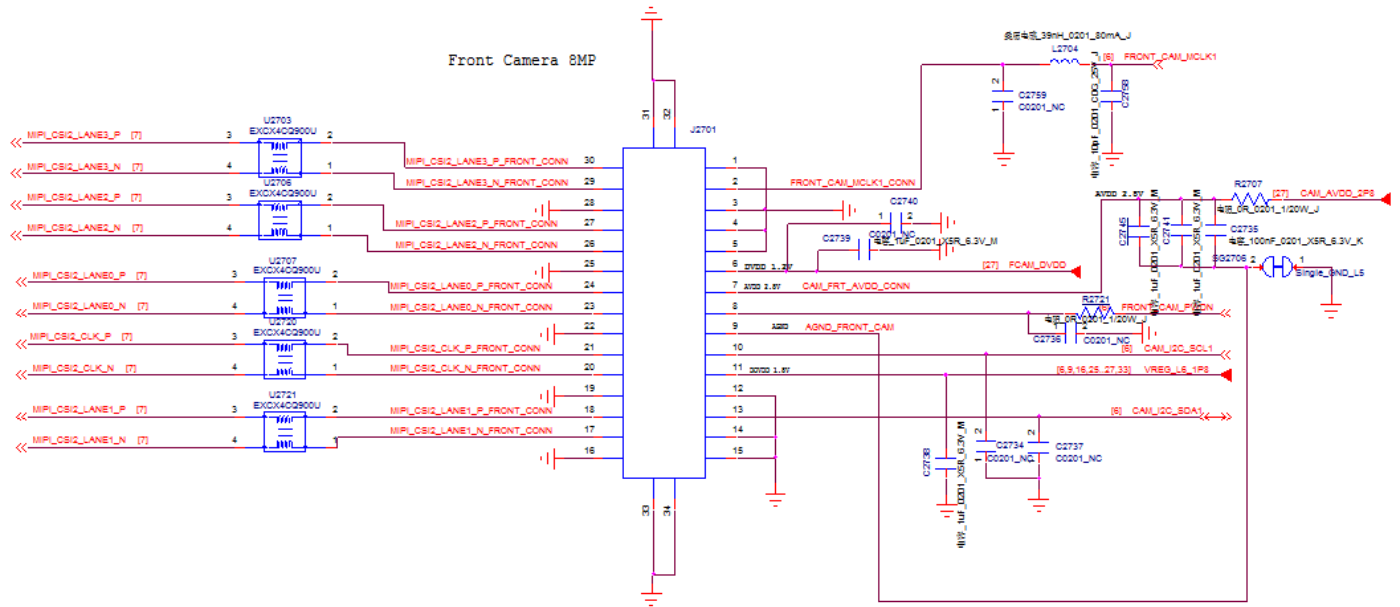
Solution:

- 1 Check camera installation for any abnormalities, re-install it or replace it before testing again;
- 2 Check the connector of the main camera J2712 for damage or soldering errors, re-solder or re-place the connector;

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- 3 Check voltage of VREG_L17_2P8 /VREG_L2_1P2 /VREG_L6_1P8;
- 4 Check U2713/U2714/U2716/ U2718/ U2719 EMI filter for damage or soldering error;
- 5 Add-solder or re-solder PMIC or re-place it;
- 6 Add-solder or re-solder CPU or re-place it;

12. Slave camera function failure

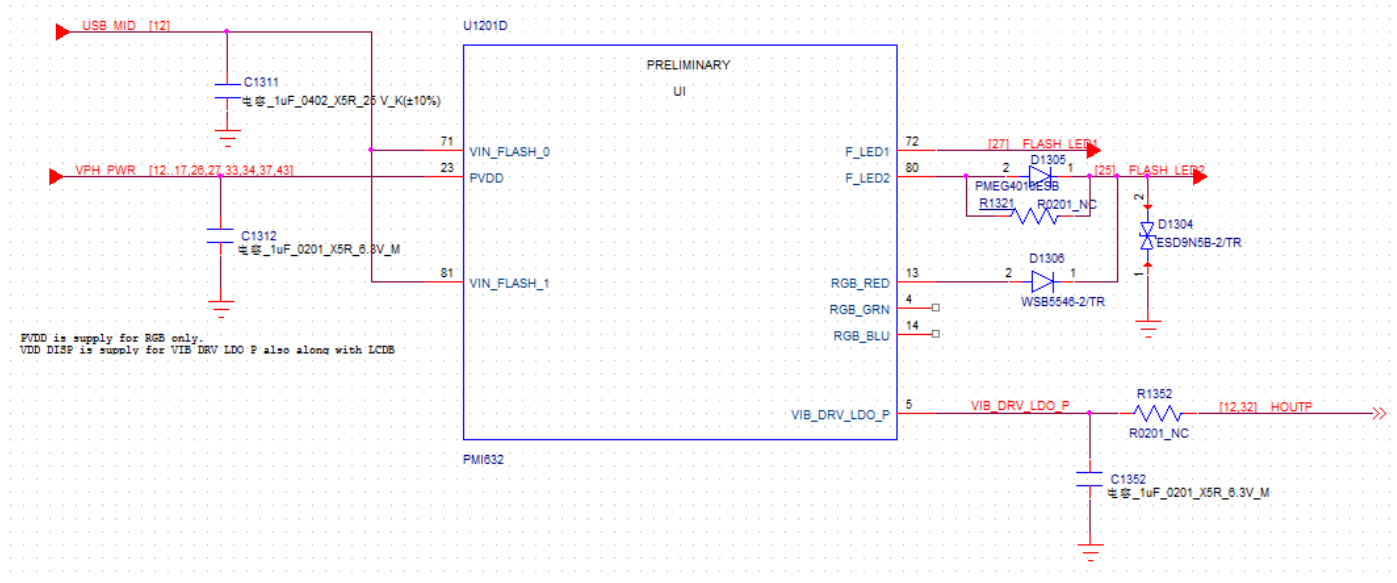


Solution:

- 1 Check camera installation for any abnormalities, re-install it or replace it before testing again;
- 2 Check the slave camera J2701 connector for damage or soldering errors, re-solder or re-place the connector;
- 3 Check CAM_AVDD / FCAM_DVDD /VREG_L6_1P8 for voltage errors, if so, check PMIC for output errors or peripheral circuits for short circuits or broken circuits;
- 4 Check U2703/U2706/U2707/U2720/U2721 EMI filter for damage or soldering errors;
- 5 Check whether the LDO chip(U2704) or its soldering is normal;
- 6 Add-solder or re-solder CPU;

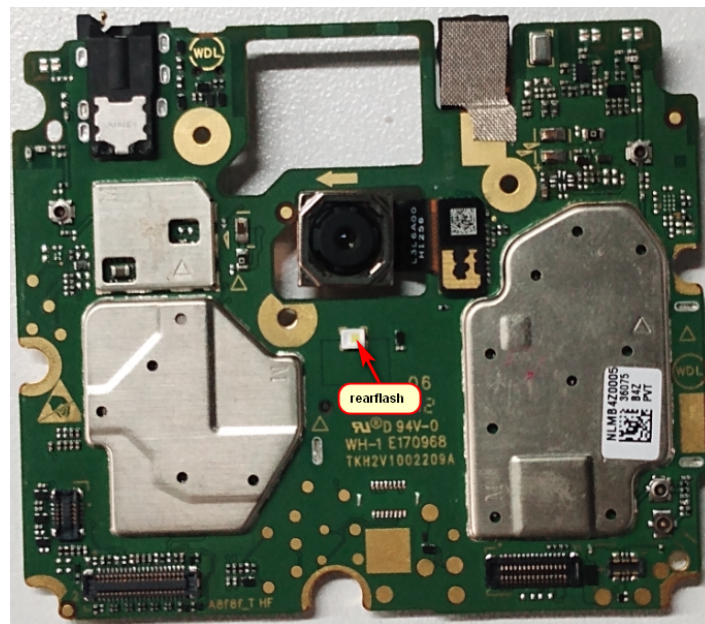
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13. Flashlight failure

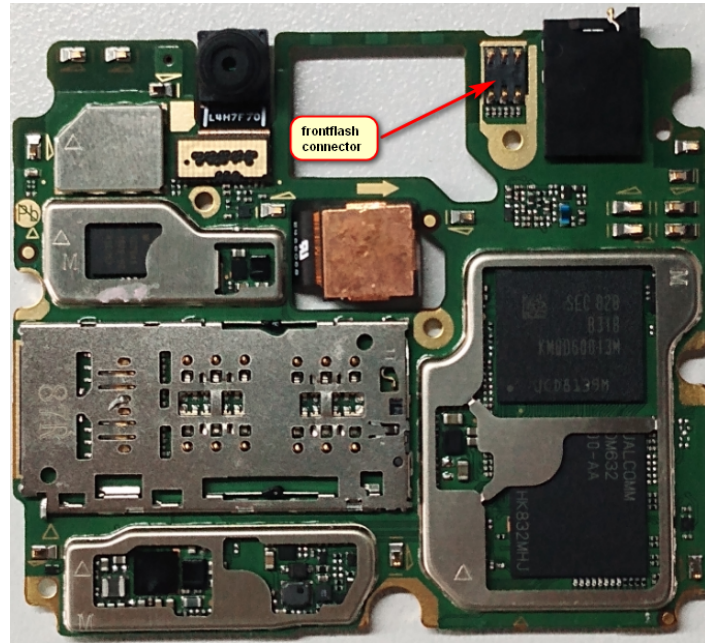


Solution:

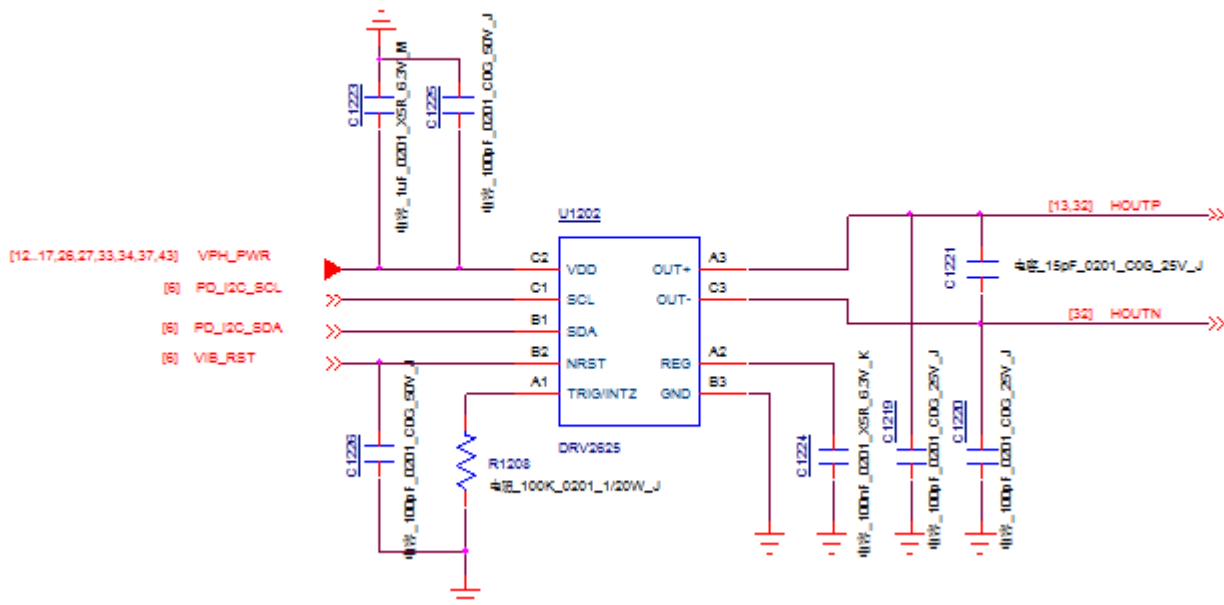
- 1 Solder or re-place U1201;



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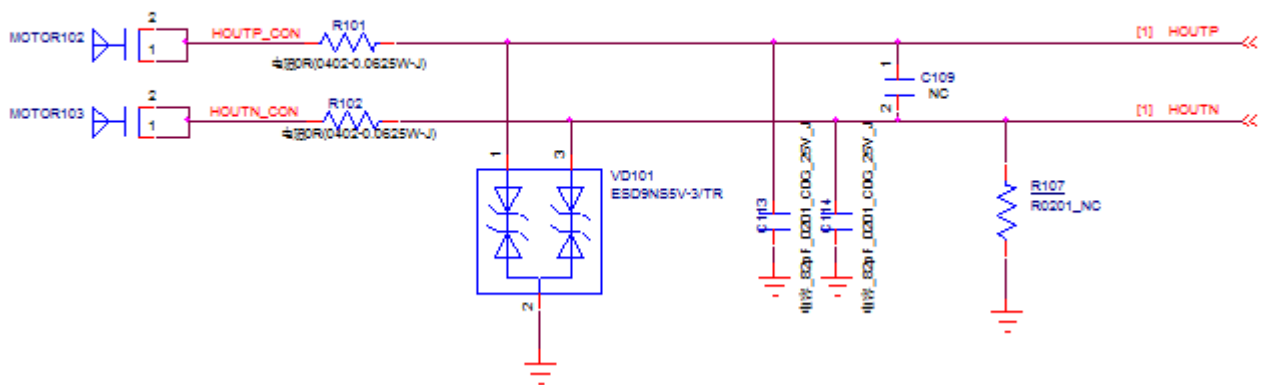


14. Motor error



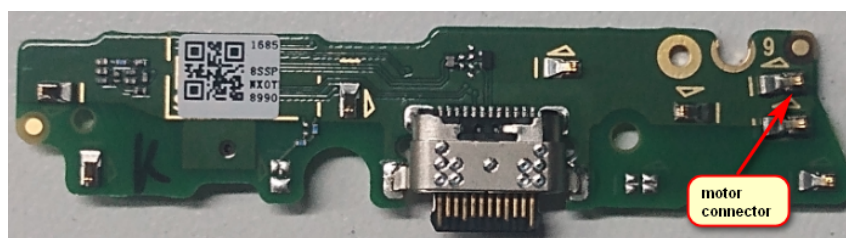
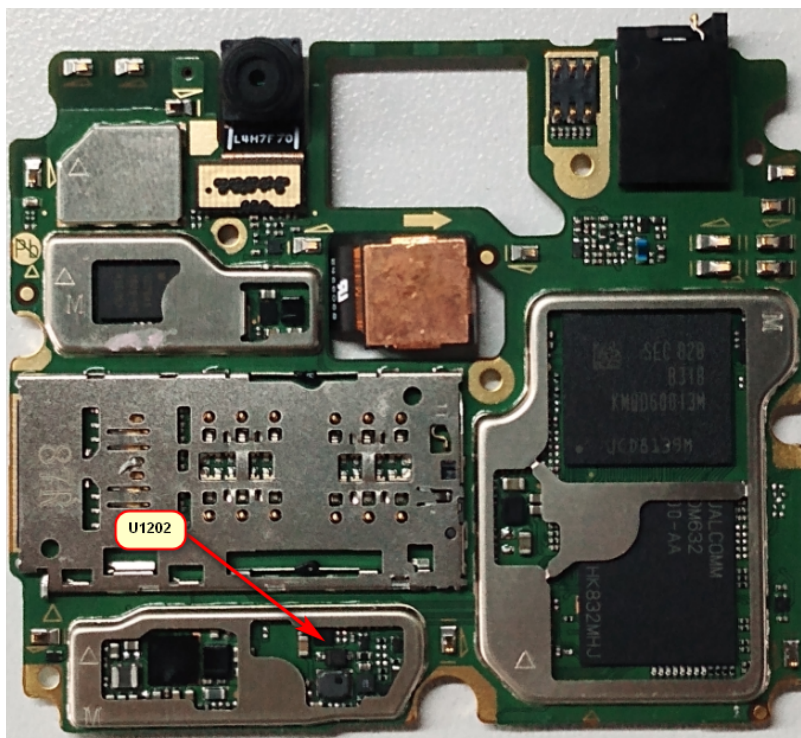
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MOTOR



Solution:

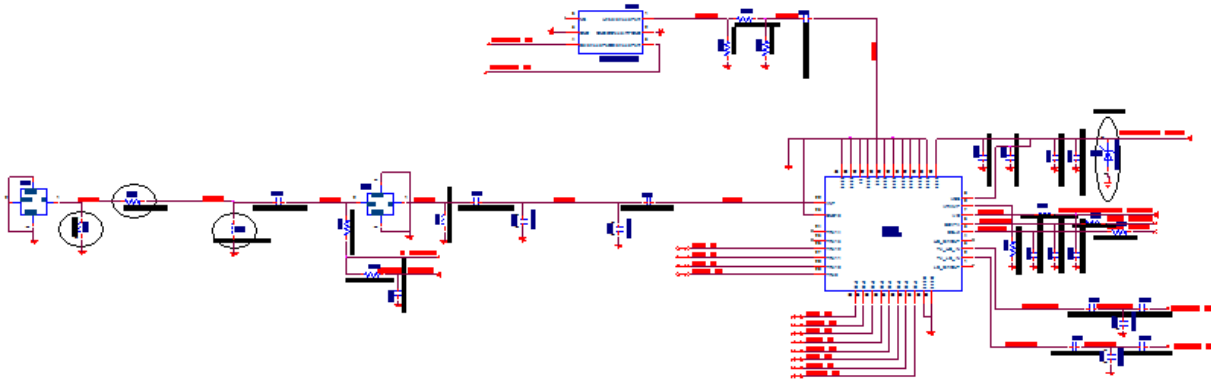
- 1 Check whether the R101, R102 is normal;
- 2 Check whether MOTOR102, MOTOR103 is normal;
- 3 Replace the motor;
- 4 Re-solder power management chip U1202 or replace it;



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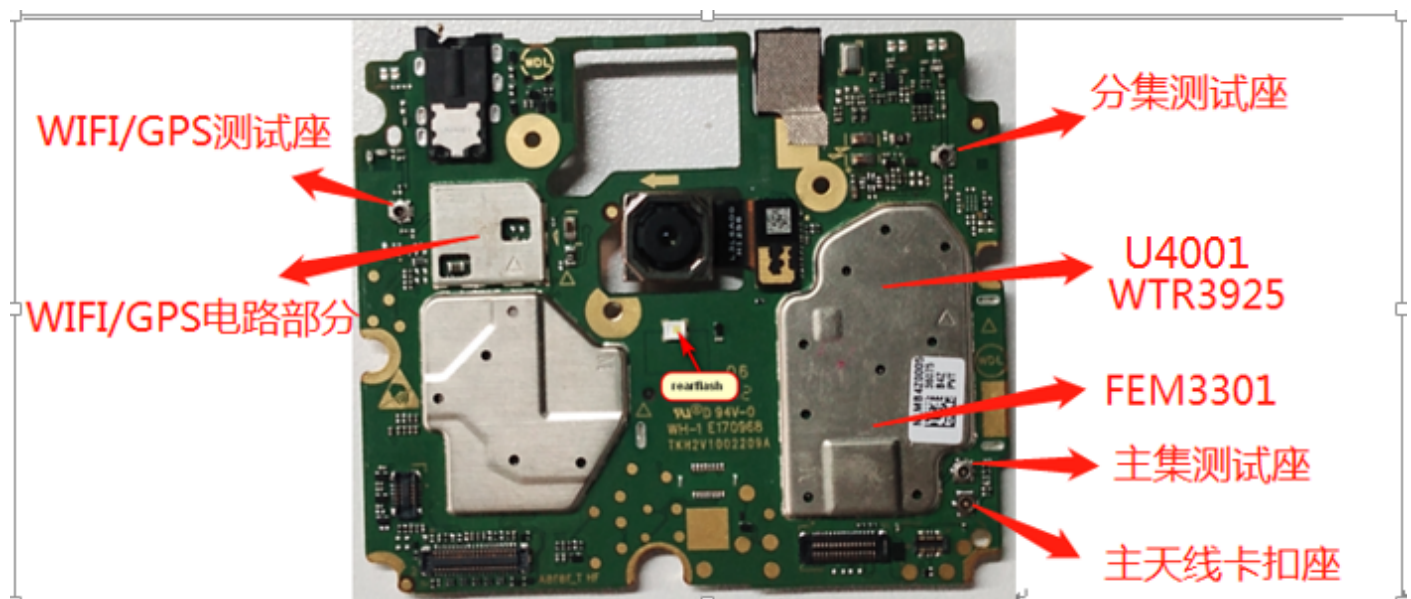
15. GSM network failure

PRIMARY_ANTENNA



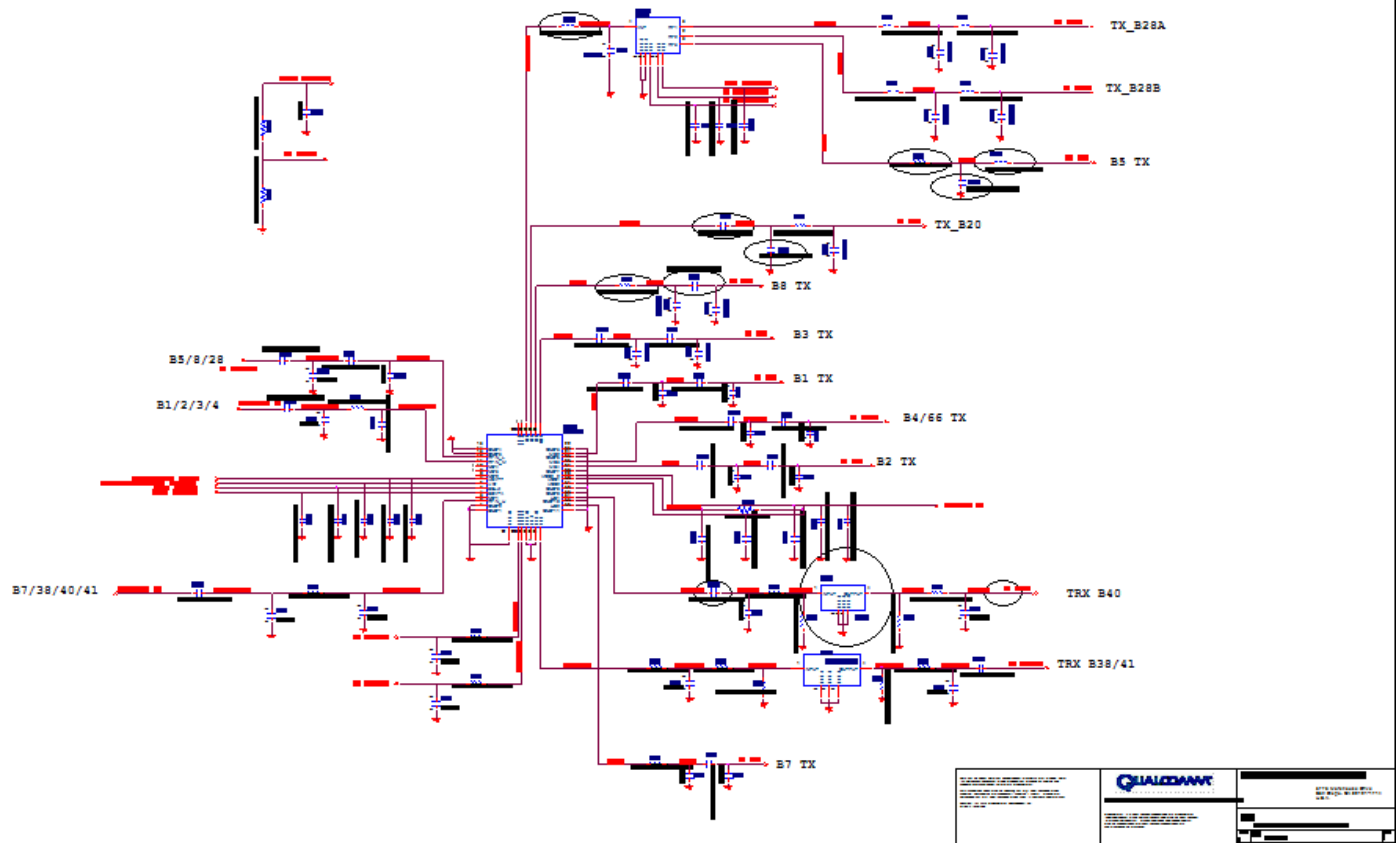
Solution:

- 1 Check PCBA antenna dome soldering status, contact with phone casing antenna or damaged body;
- 2 Check the connection of RF connector and circuit components, Inductors, resistors for soldering errors after collision;
- 3 Check front RF switch FEM3301 and relevant components for soldering errors and body damage;
- 4 Re-calibrate motherboard RF indicators. If it fails, troubleshoot and repair it for failed bands;
- 5 Re-solder or replace FEM3301 RF transceiver chip;
- 6 Re-solder or re-place U4300;
- 7 Re-solder or re-place U4001;
- 8 Re-solder or re-place U401;



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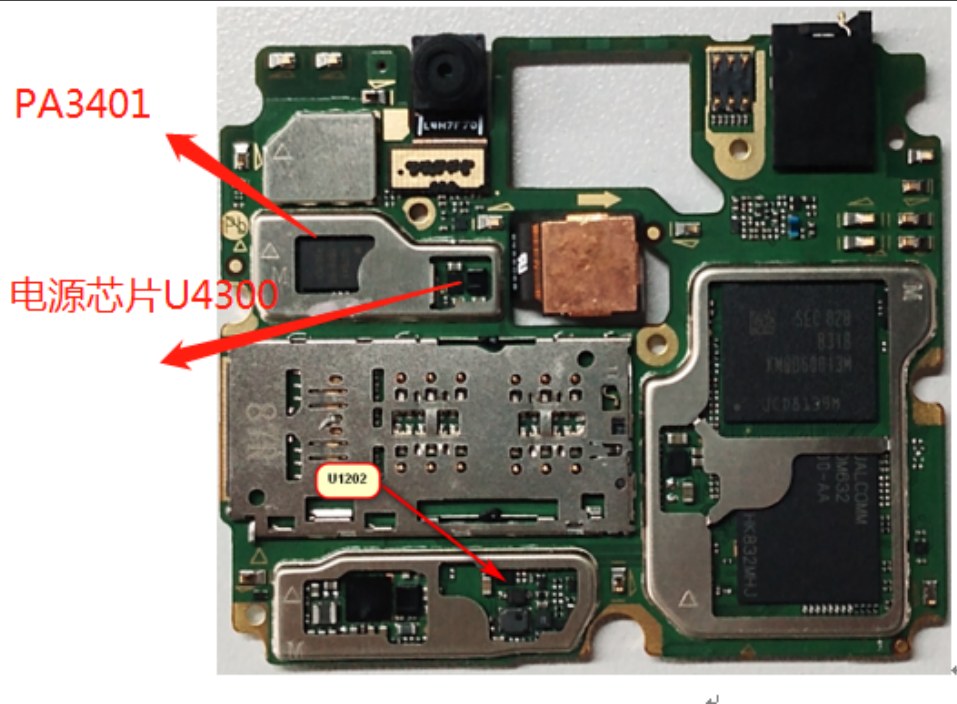
16. WCDMA and LTE network failure



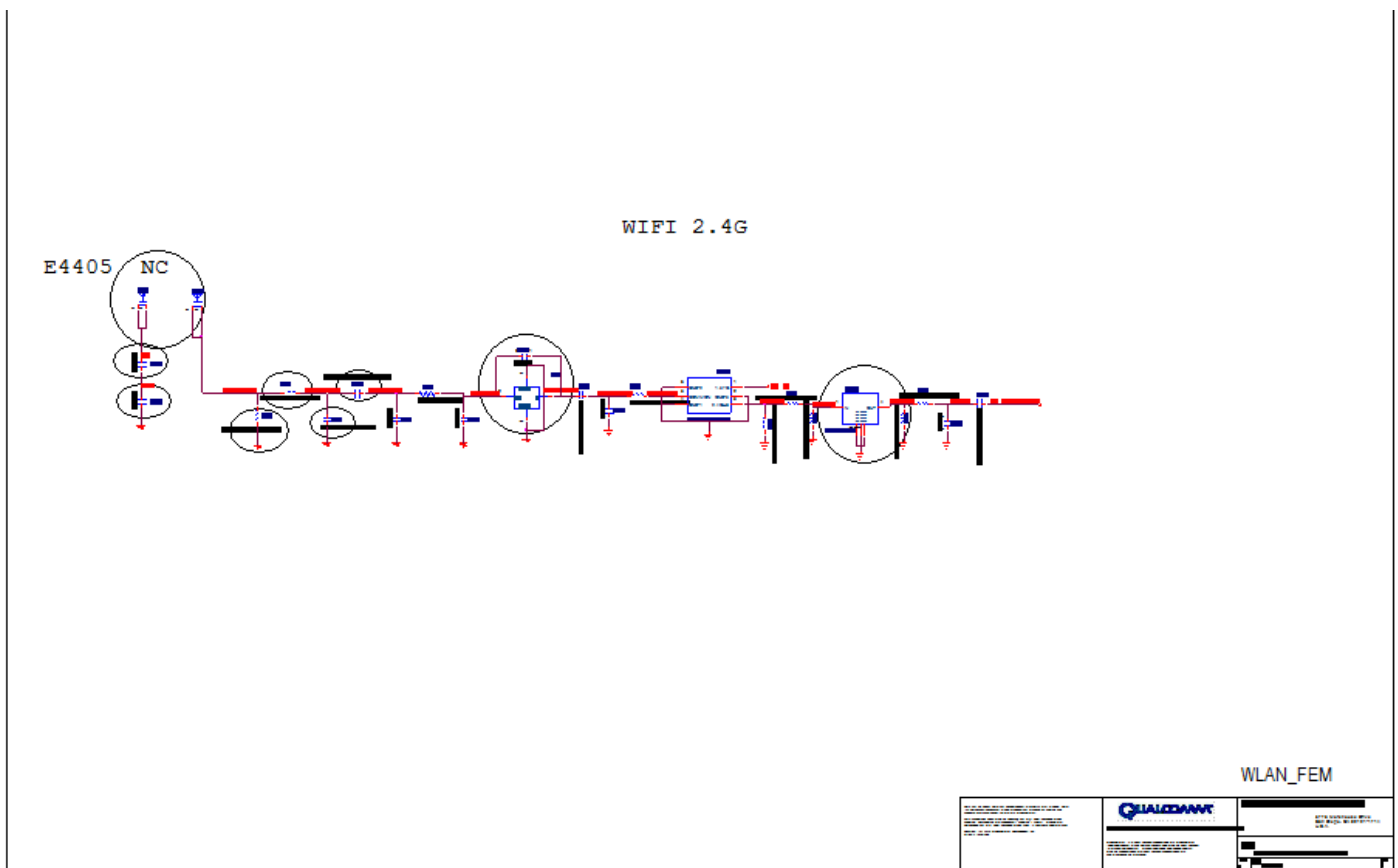
Solution:

- 1 Check whether main and branch set test seat is falsely soldered;
- 2 Calibrate phone MB, pinpoint cause of error, determine receiving or transmitting errors according to bands with errors, troubleshoot relevant circuit components;
- 3 Add-solder or replace FEM3301, PA3401, 4300, duplex DU, Quad QP, Filter FL, RF Transceiver U4001;
- 4 According to RF indicators, check components in the circuit at the corresponding band for soldering errors or body damage;
- 5 Troubleshoot relevant band of diversity filter module according to source drawing;
- 6 Re-solder or re-place U401 CPU;

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17. WIFI/Bluetooth/FM

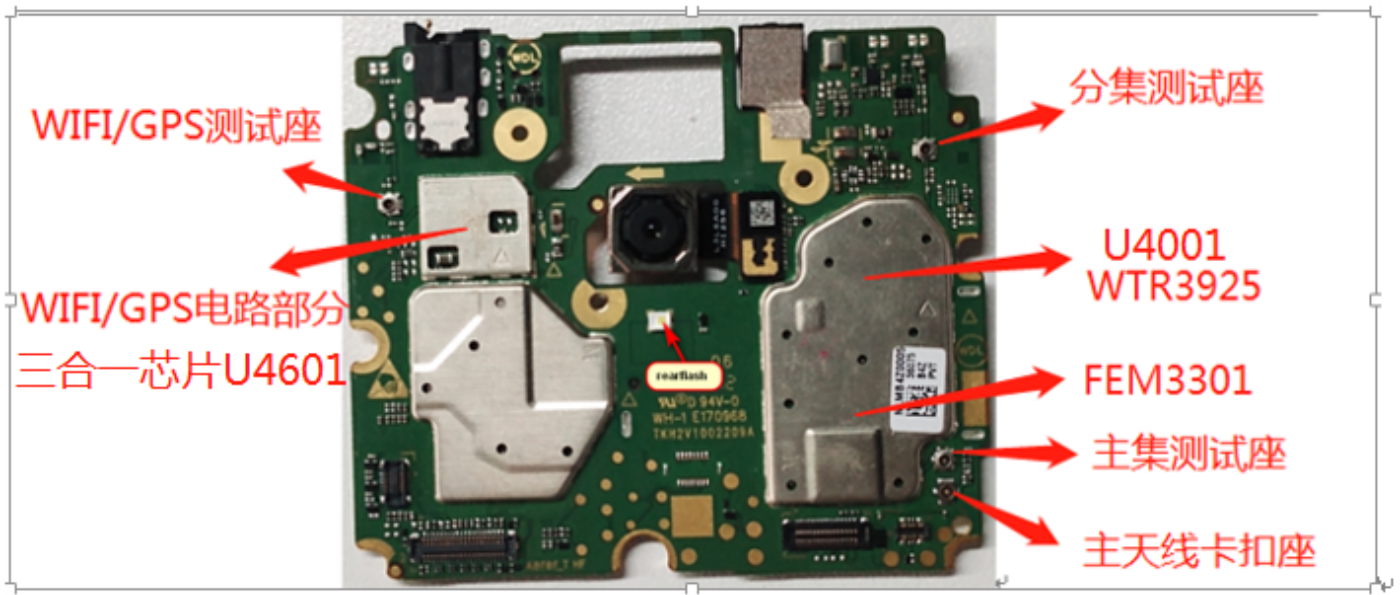


Solution:

- 1 Check MB's WIFI/BT antenna channel components or their soldering for damage;
- 2 Check the RF test stand for false solder joints;
- 3 Check headset channel connection, FM antenna circuit in headset;
- 4 Check filter FL4401, DI4401 for soldering offset or body damage;

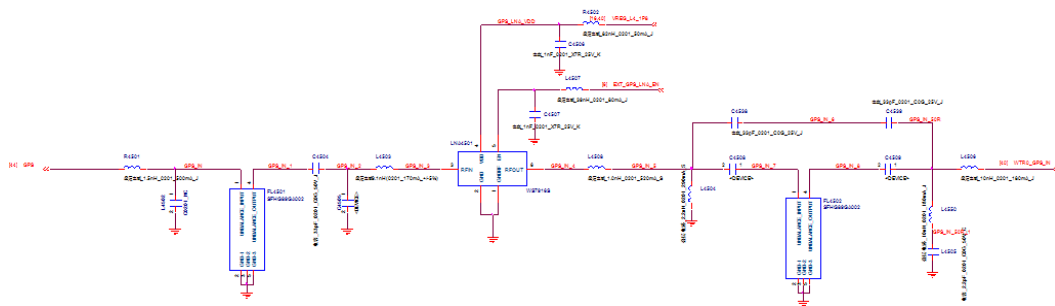
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- 5 Add-solder or re-solder WLAN/BT/FM 3-in-1 chip or re-place it before testing again;
6 Add-solder or re-solder CPU U401, or re-place it before testing it again;



18. GPS failure

GPS

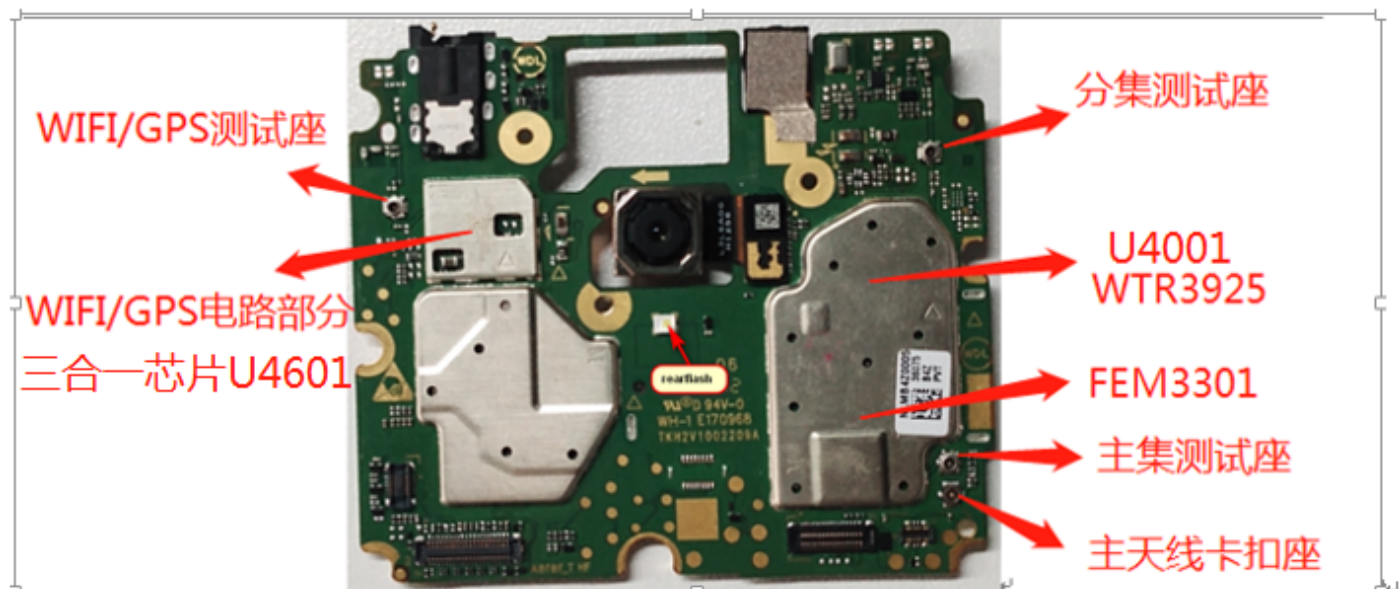


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Solution:

- 1 Check GPS channel components for soldering or body errors;
- 2 Check the RF test stand for false solder joints;
- 3 Check DI4401 for soldering offset or body damage;
- 4 Check filter FL4501, FL4502 for soldering or body errors;
- 5 Check LNA chip LNA4501 for soldering or body errors;
- 6 Add-solder or re-solder U4001 chip or re-place it before testing again;
- 7 Add-solder or re-solder CPU U401, or re-place it before testing it again;

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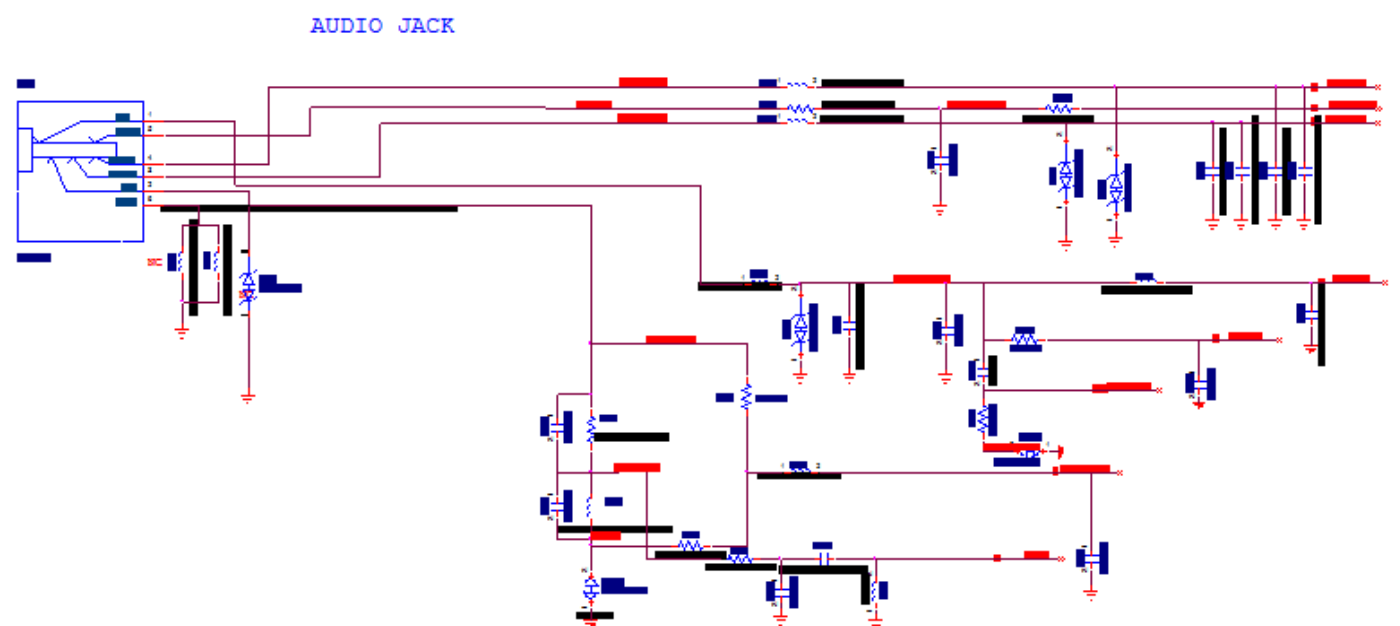


19. Self-calibration (SELF_CAL) and XO calibration failure:

Solution:

- 1 Check whether 5 mipi communication are OK, if so, identify errors for individual analysis or take the following steps:
- 2 Check whether U4001 is of false solder joints, continuous tin or poor quality; Re-solder or re-place U4001;
- 3 Check whether FEM3301 is of false solder joints, continuous tin or poor quality; Re-solder or re-place FEM3301;
- 4 Check whether the U4001 is of false solder joints, continuous tin or poor quality; Re-solder or re-place U4001;
- 5 Check whether the G1401 is of false solder joints, continuous tin or poor quality; Re-solder or re-place G1401;
- 6 Check whether the U1401 is of false solder joints, continuous tin or poor quality; Re-solder or re-place U1401;

20. Headset identification failure/headset MIC errors

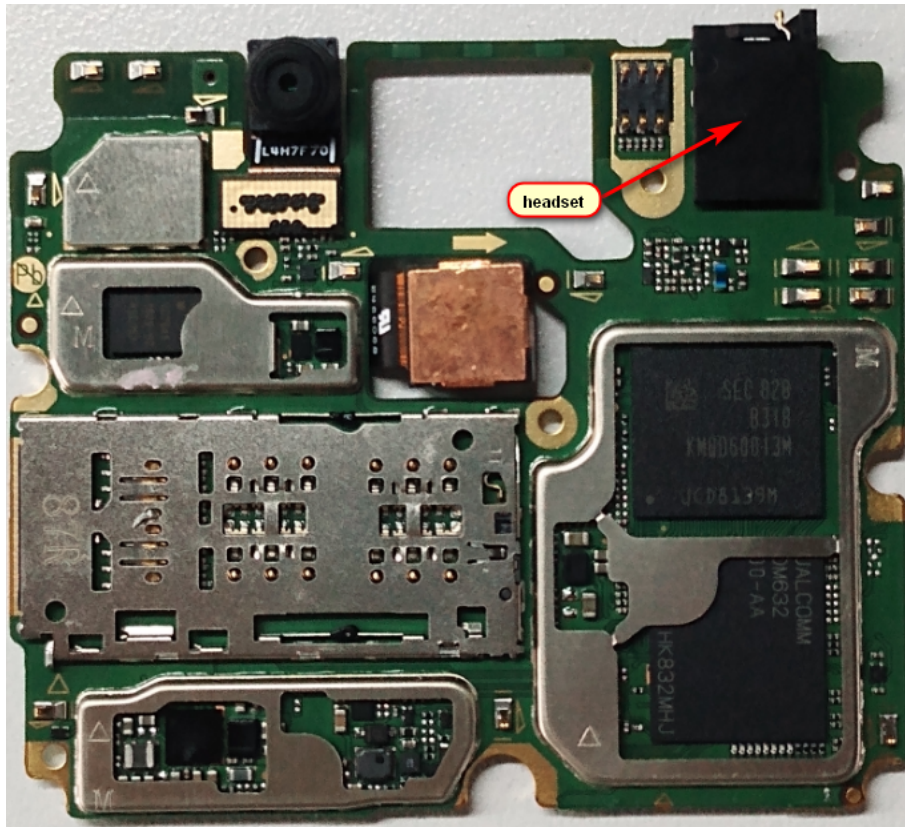


Solution:

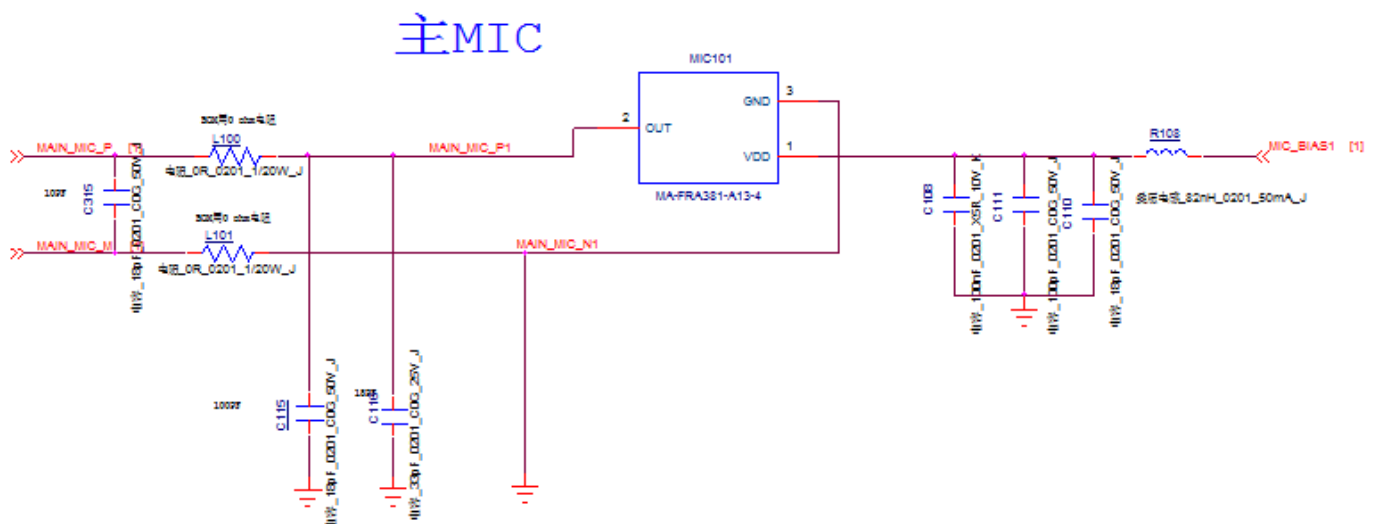
- 1 Re-test headset function after replacing headset;

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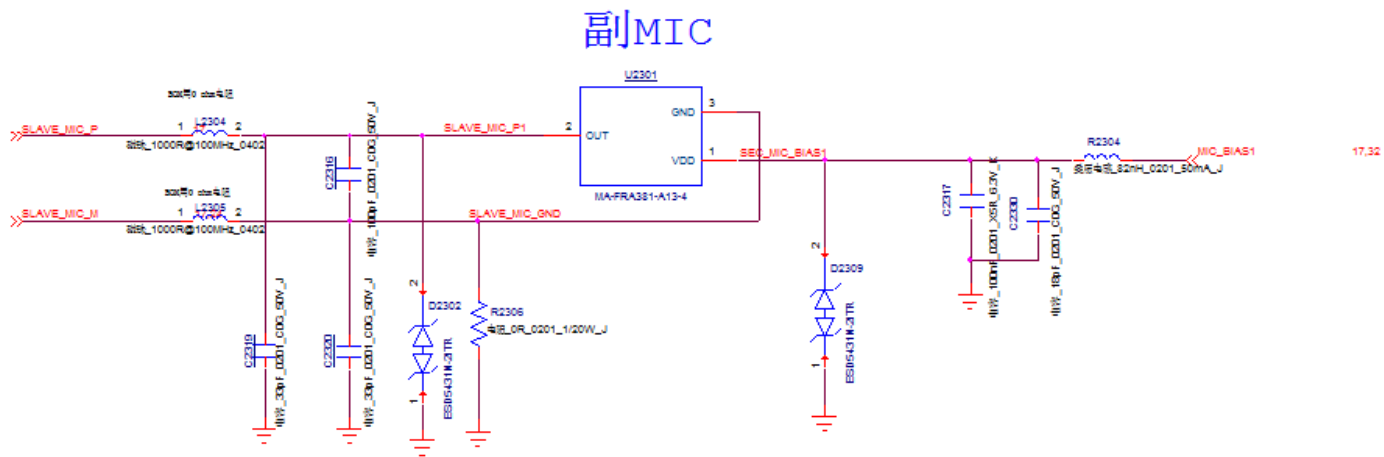
- 2 Check headset jack for damage or soldering for skews or over height, re-solder or replace headset jack;
- 3 Check receiver circuit components for damage or soldering for errors, repair before testing it again;
- 4 Re-solder or re-place codec U1801;



21. MIC silent/low volume

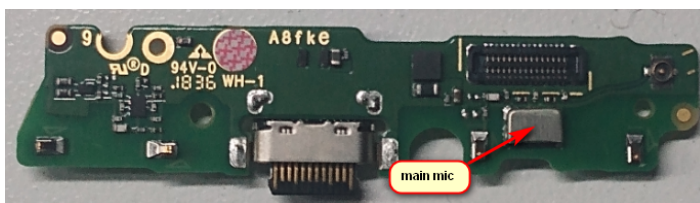
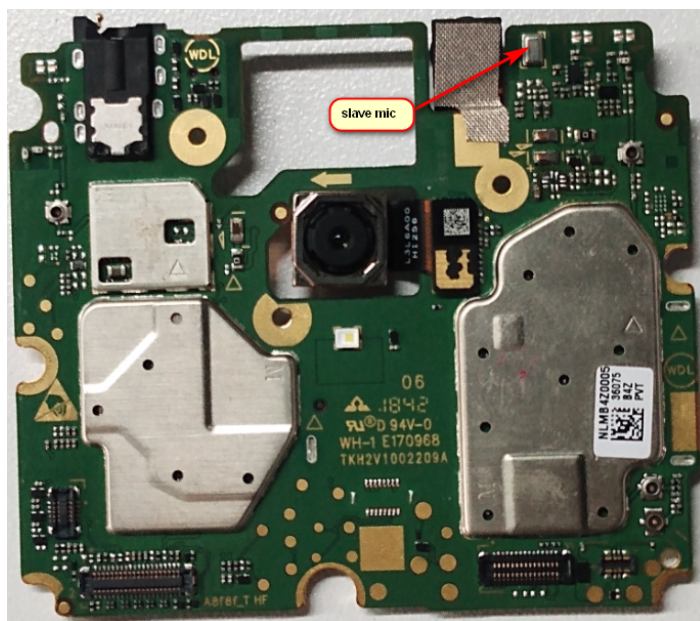


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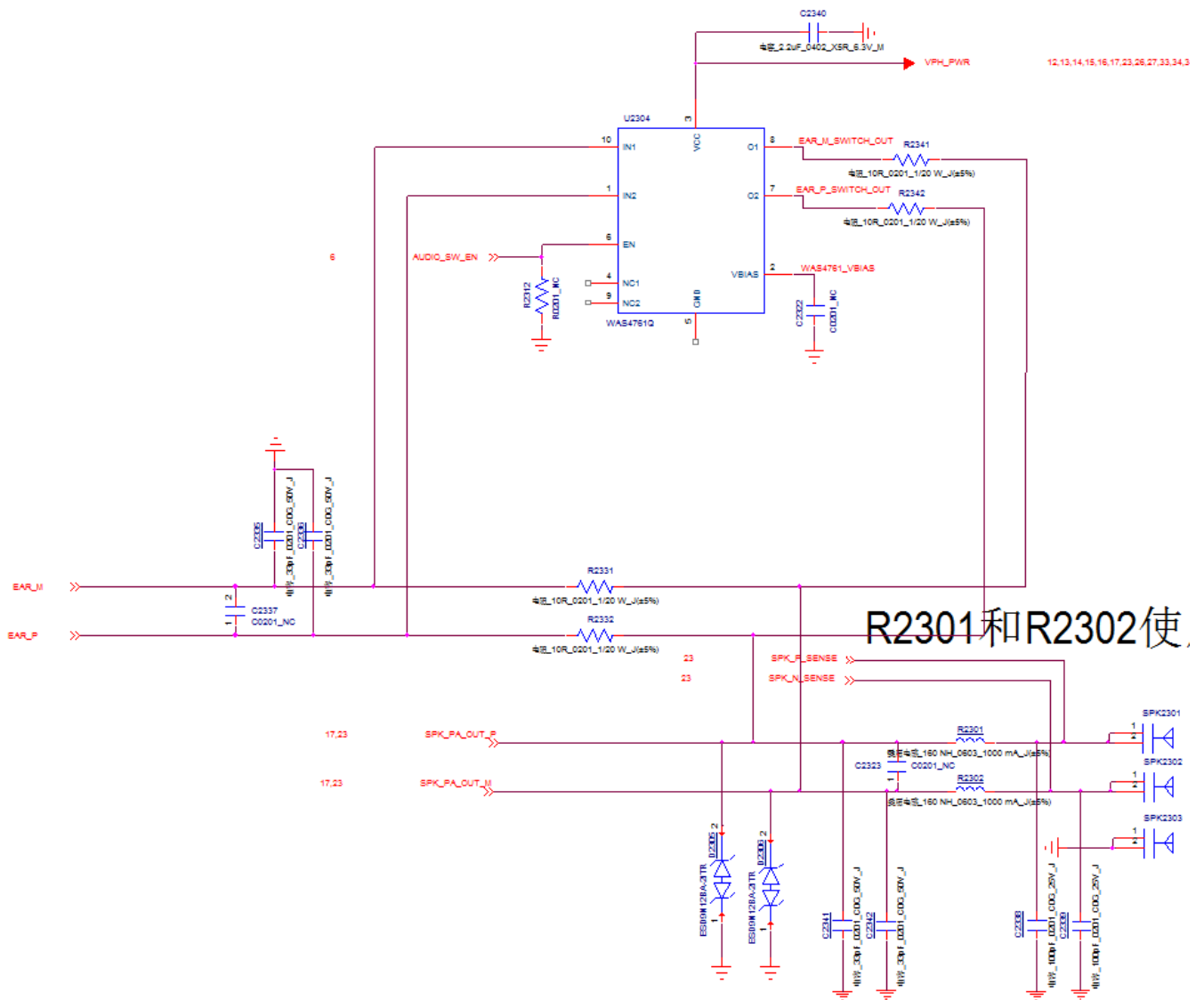
Solution:

- 1 Visually check primary and secondary MIC relevant components for soldering offset or body damage;
- 2 Re-place primary and secondary MIC before testing it again. Pay attention to high temperature during replacement as it may damage it.
- 3 Add-solder or re-solder U1201, or re-place it before testing again;



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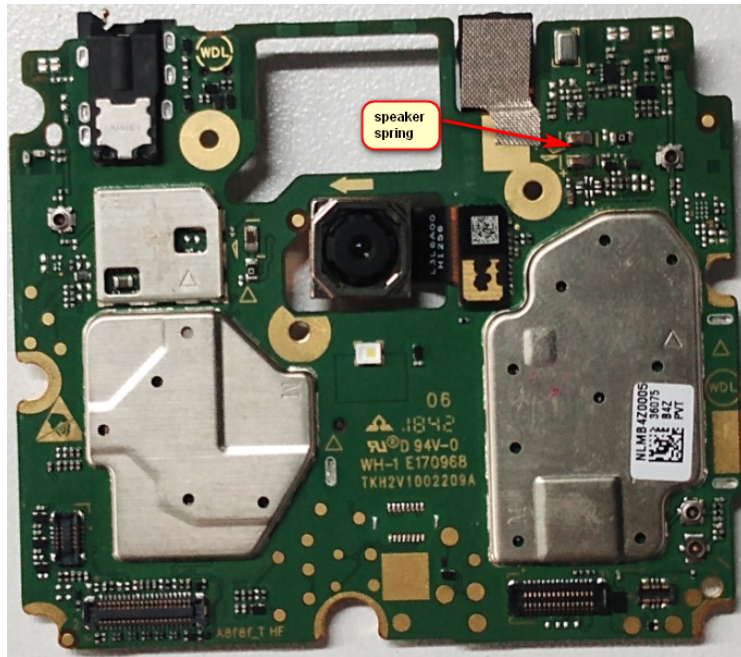
22. Speaker silent



Solution:

- 1 Visual check relevant components for damage or soldering error;
- 2 Check the MB dome and speaker FPC for poor soldering including skew and continuous tin;
- 3 Re-solder or re-place U2304 or smart PA U2311.

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23. IF station FASTBOOT_FLASH_01

Solution:

- 1 Check whether the earth voltage of VREF_LPDDR3 is normal (0.6V or so).
- 2 X-ray U401 to check for continuous tin
- 3 Check the resistances around CPU and check R404

24. IF station WAIT_FOR_ALL_INTERFACE_ARRIVAL_01

Solution:

- 1 Power on the USB and check whether the current of the motherboard is too large. If more than 100mA, find the power supply that is short-circuited
- 2 Measure the resistance of USB_HS_DP, USB_HS_DM to check whether the earth impedance is abnormal;
- 3 Check U101 for soldering errors;